

2022

COMPUTER SCIENCE AND ENGINEERING

Paper : CSEL-1013

(Computational Geometry)

Full Marks : 70

The figures in the margin indicate full marks.

*Candidates are required to give their answers in their own words
as far as practicable.*

Answer **Question No. 1** and **Question No. 2** and **any four** from the rest.

1. Define and explain **any five** from the following : 2×5
 - (a) Visibility graph
 - (b) y-Monotonicity of a polygon
 - (c) Boundedness
 - (d) Left-hand turn
 - (e) Plane-sweep algorithm
 - (f) Order reversing
 - (g) Spanner property.

2. Answer **any five** from the following : 4×5
 - (a) Define and explain the terms *helper* and *fix-up* with respect to the polygon triangulation problem, with the help of necessary diagrams.
 - (b) Define the *line segment intersection problem* and the term *event(s)* relating to this problem. Briefly explain how the events are processed following some algorithm for this problem.
 - (c) State the problem of computing *guarding points* of an art gallery, and comment about its upper limit for an art gallery of n sides. In this context, establish the linkage to discrete combinatorial geometry.
 - (d) Discuss a generalized algorithm to find how a monotone polygon can be triangulated.
 - (e) Define the half-plane intersection problem, and show how this problem can be utilized to compute the convex hull for a given set P of n points on the plane.
 - (f) Write a short note on *computing slope statistics*.
 - (g) What do you mean by *maximizing angles* and *edge flipping* relating to Delaunay triangulations?

Please Turn Over

3. Clearly compare the complexities of computing convex hull by the methods of *divide-and-conquer* and *gift-wrapping*, and justify the complexities by briefly describing the underlying algorithms. 3+7
 4. What do you mean by trapezoidal map? Devise an algorithm to compute the same for a given set of line segments in the plane. In this respect, clearly mention the basic assumptions and the fundamental principle of the algorithm. Also, measure the time complexity of the algorithm devised by you. 2+6+2
 5. Define the *d-dimensional linear programming problem*. How may such a problem instance provide a feasible polytope, or the other alternatives? Briefly describe an algorithm that intersects two convex polygons into a single convex polygon, and also quantify its time complexity. 2+3+5
 6. Define and explain upper envelope as well as lower envelope relating to half-plane intersection problem. Briefly describe how these envelopes are related to realizing a convex hull following the theory of point-line duality. 4+6
 7. (a) State and prove the *empty-circle property* and *closest pair property* in order to compute Delaunay triangulation.
(b) Consider the following three line equations in the primal plane :
$$l_1 : 2x - y = -1; \quad l_2 : x + 2y = 12; \quad l_3 : -5x + 3y = 5.$$

Comment about the points l_1^* , l_2^* and l_3^* in the dual plane, and justify. 5+5
 8. Define and explain the usefulness of *beach line* relating to the computation of Voronoi diagram for a given set of sites in the plane. Following the *Fortune's algorithm*, describe the chronological development of a beach line till it disappears starting from its birth. Also, analyze the time complexity of the algorithm. 3+5+2
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